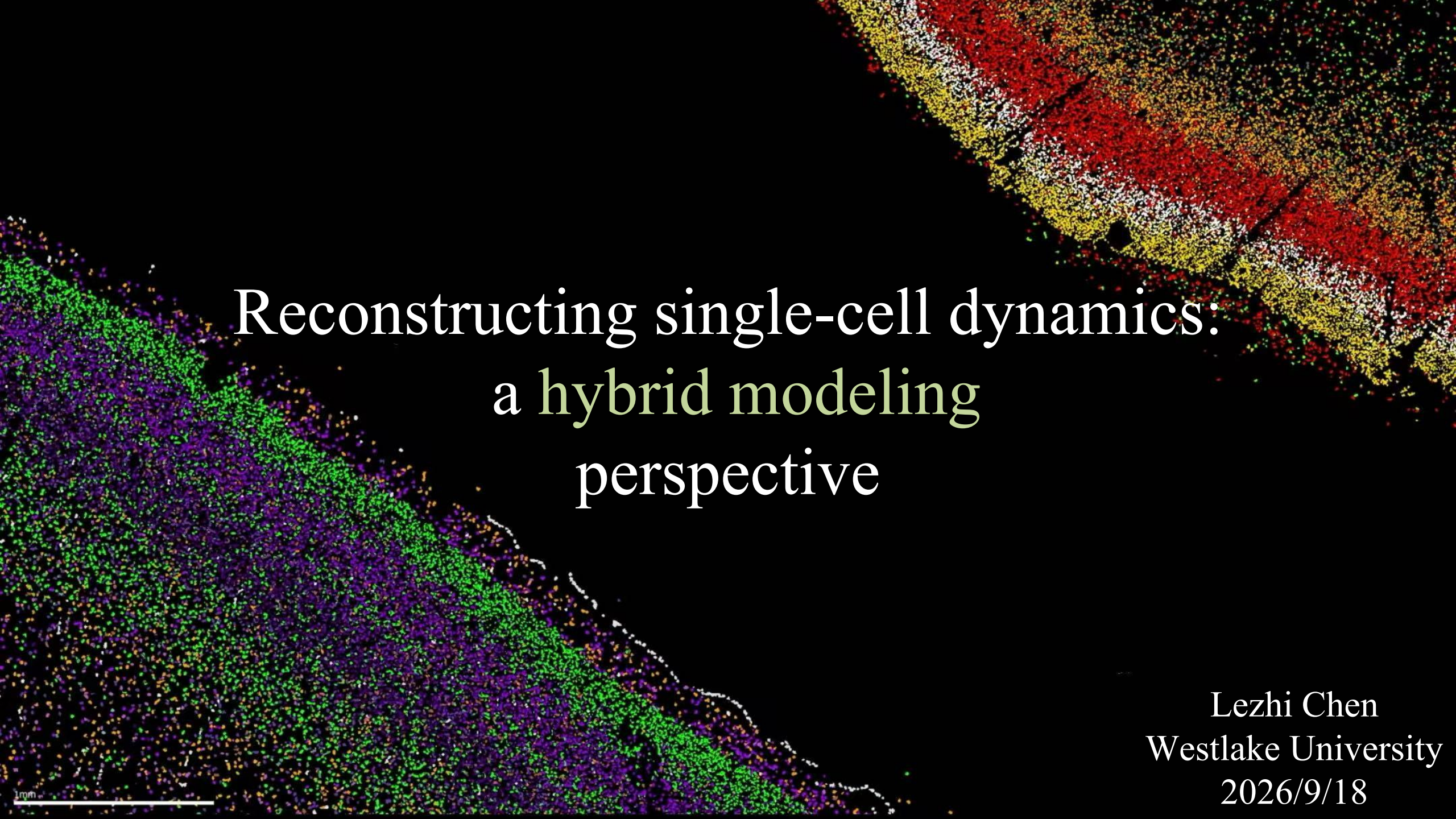
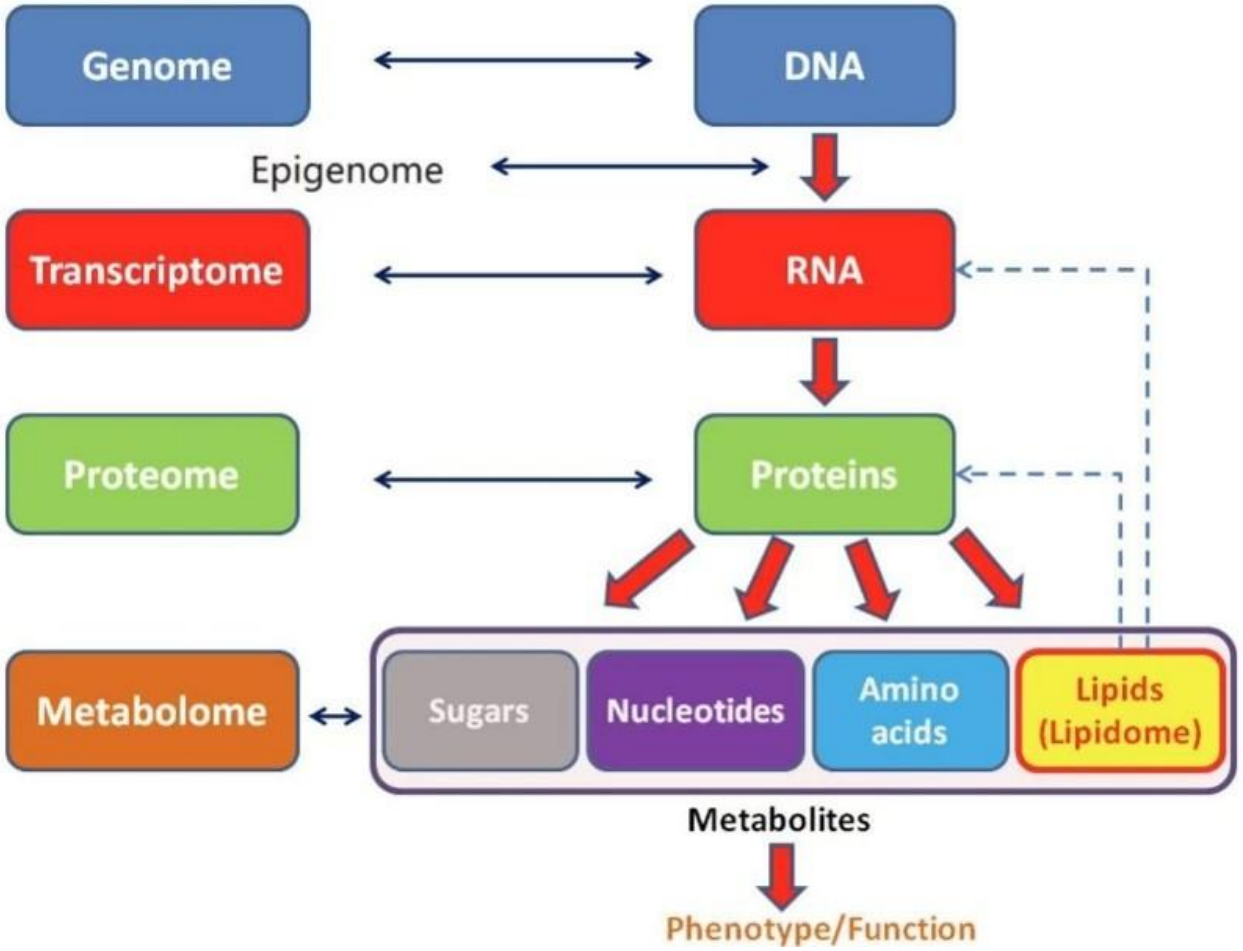



# Reconstructing single-cell dynamics: a hybrid modeling perspective

Lezhi Chen
Westlake University
2026/9/18

# Modeling living systems

mechanistic modeling v.s. data-driven modeling

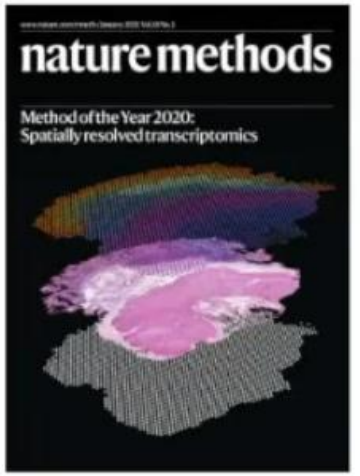



$$\frac{\partial \rho}{\partial t} = D_\rho \nabla^2 \rho - \chi \nabla \cdot (\rho \nabla c)$$

$$\frac{\partial c}{\partial t} = D_c \nabla^2 c + \alpha \rho - \beta c$$

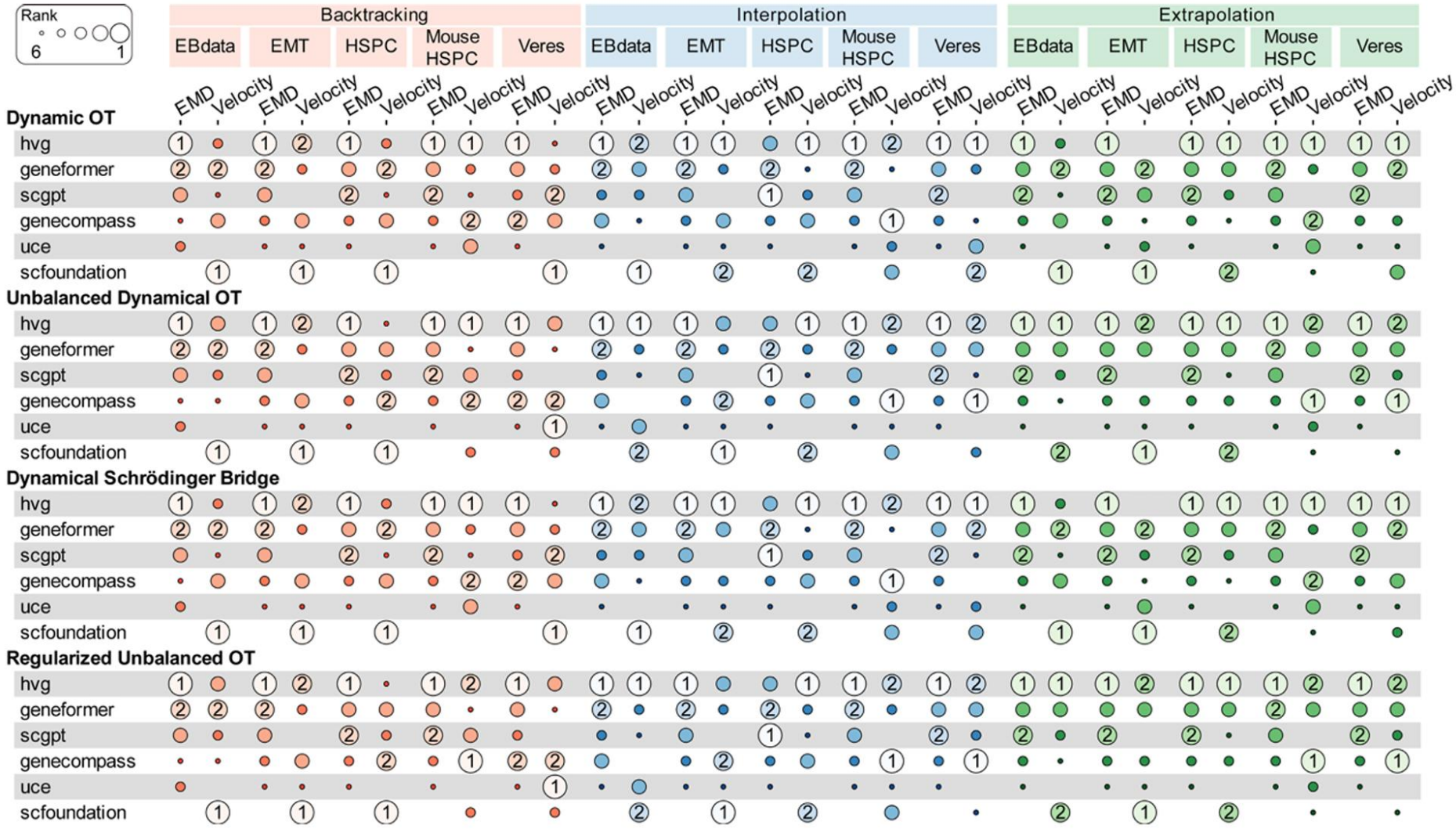



scRNA-seq (2018)




Spatial transcriptomics (2020)

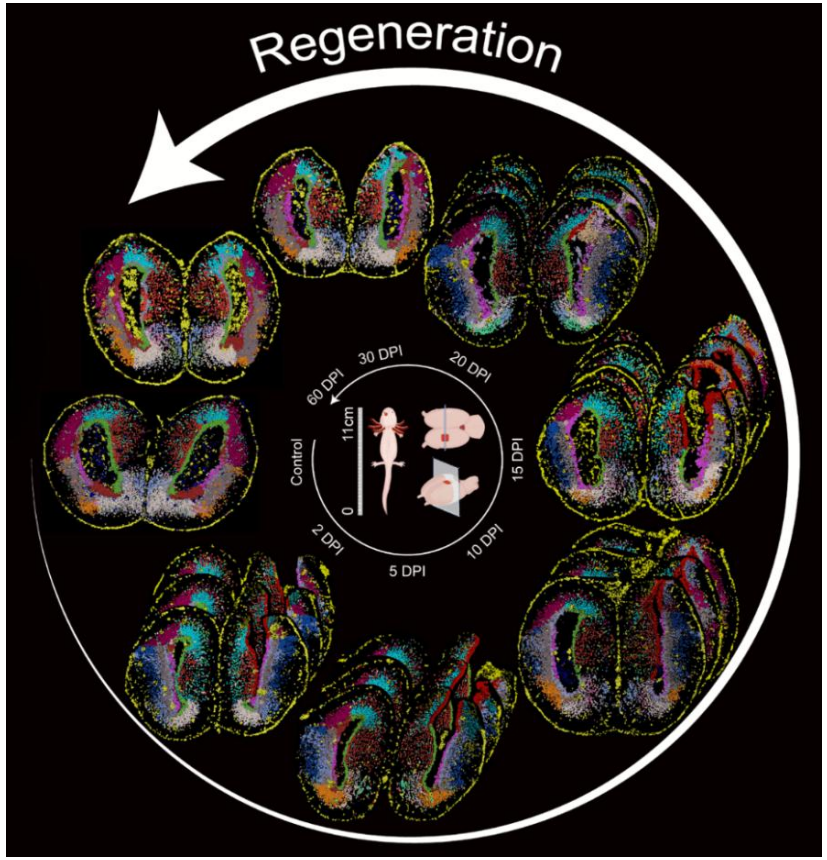
# But data-driven methods do not always work !



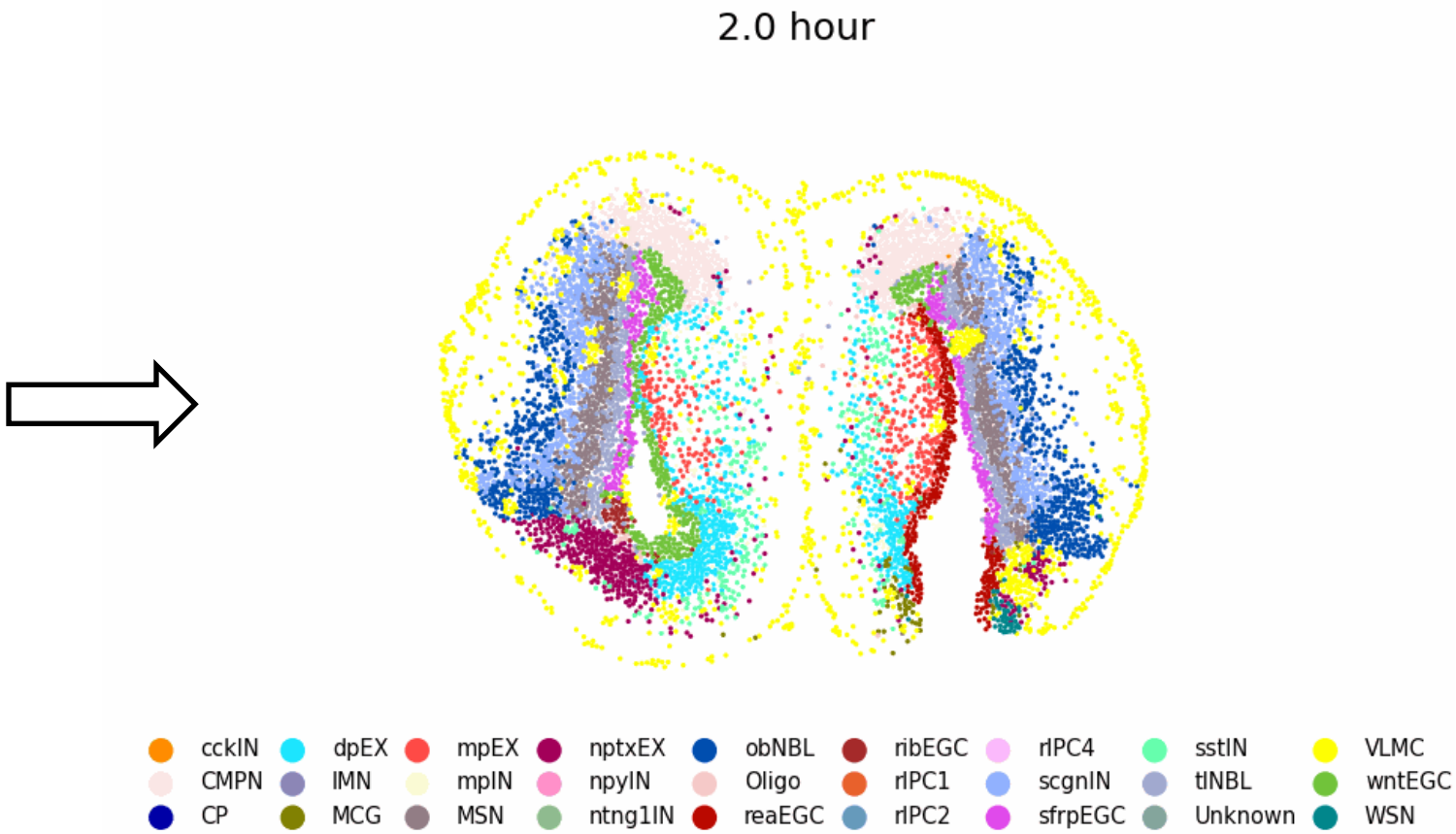

Zhou, et al. bioRxiv 2026

**Why not combine the strengths of both ? That's exactly what hybrid modeling is about !**

# Single-cell spatiotemporal dynamics reconstruction

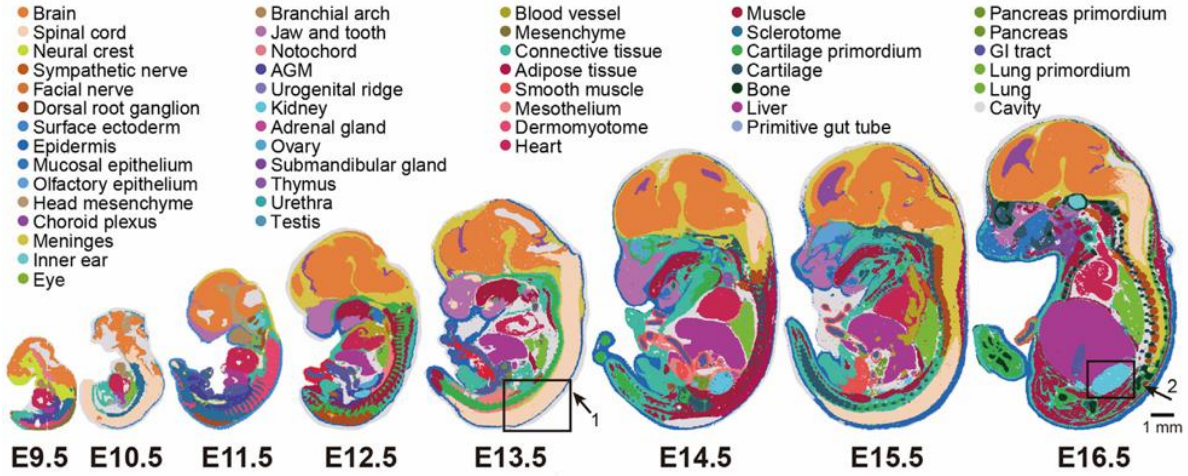



Wei, X, et al. Science 2022




Peng, Zhou, ... Nature Methods 2026

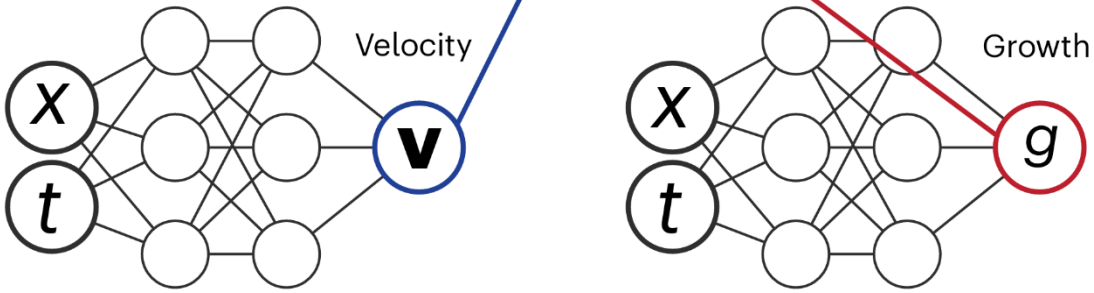
# Dynamical systems modeling

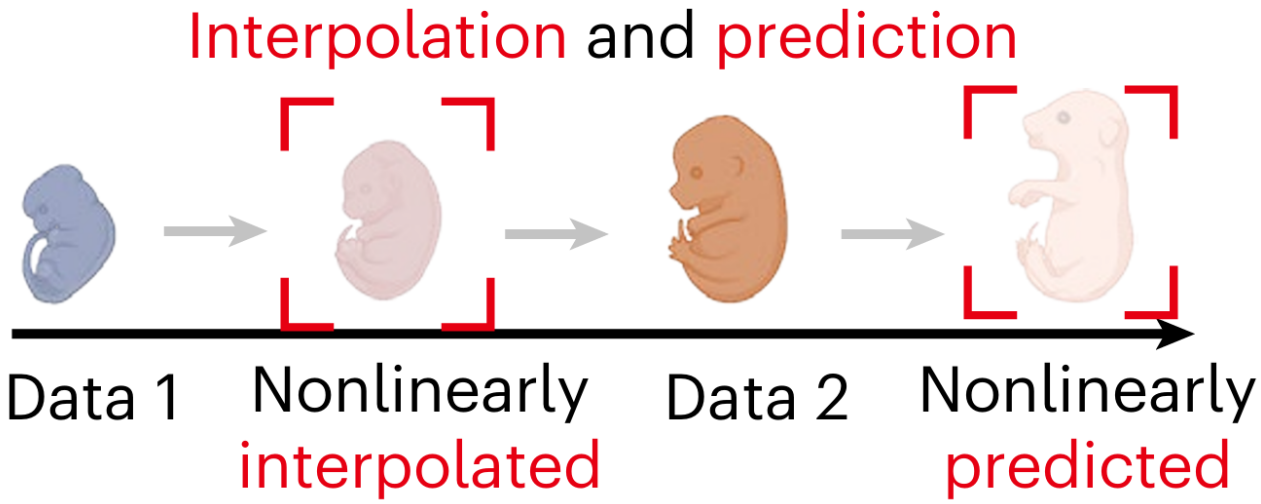



$$\frac{d}{dt}\begin{pmatrix}\boldsymbol{\phi}_t(\boldsymbol{x})\\ \ln m_t(\boldsymbol{x})\end{pmatrix} = \boxed{\begin{pmatrix}\boldsymbol{u}_t(\boldsymbol{\phi}_t(\boldsymbol{x}))\\ g_t(\boldsymbol{\phi}_t(\boldsymbol{x}))\end{pmatrix}}$$

$$\partial_t \rho_t(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} \cdot (\boldsymbol{u}_t(\boldsymbol{x})\rho_t(\boldsymbol{x})) = g_t(\boldsymbol{x})\rho_t(\boldsymbol{x})$$

$$\partial_t \rho + \nabla \cdot (\mathbf{v}\rho) = g\rho$$



# Optimal Transport: coupling between unpaired cells

$$\mathrm{WFR}_\delta^2(\mu_0,\mu_1)=\inf_{\rho,g,\boldsymbol{u}}\int_0^1\int_{\mathcal{X}}\frac{1}{2}(\|\boldsymbol{u}(\boldsymbol{x},t)\|_2^2+\delta^2\|g(\boldsymbol{x},t)\|_2^2)\rho_t(\boldsymbol{x})\mathrm{d}\boldsymbol{x}\mathrm{d}t$$

$$\text{s.t.}\qquad \partial_t\rho+\nabla_{\boldsymbol{x}}\cdot(\rho\boldsymbol{u})=\rho g,\ \rho_0=\mu_0,\ \rho_1=\mu_1,$$

**traveling Dirac** Liero et al. Inventiones mathematicae 2018

$$\mathrm{WFR\text{-}DD}_\delta^2(m_0\delta_{\boldsymbol{x}_0},m_1\delta_{\boldsymbol{x}_1})=2\delta^2(m_0+m_1-2\sqrt{m_0m_1}\overline{\cos}(\frac{\|\boldsymbol{x}_0-\boldsymbol{x}_1\|_2}{2\delta}))$$

$$m(t)=At^2-2Bt+m_0,\quad \boldsymbol{u}(t)m(t)=\boldsymbol{\omega}_0$$

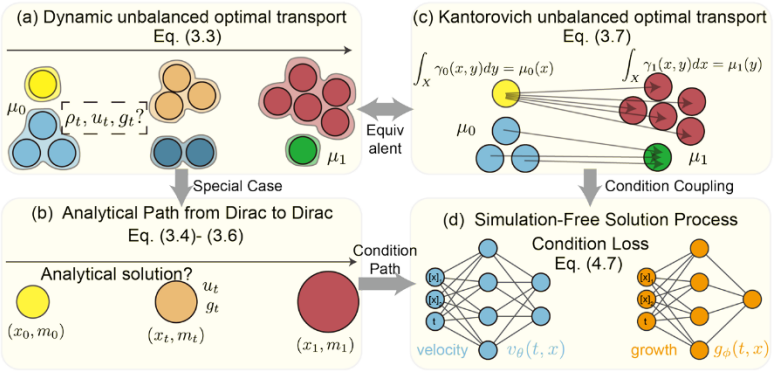
$$\begin{cases}A=m_0+m_1-2\sqrt{\dfrac{m_0m_1}{1+\tau^2}}\ ,\ B=m_0-\sqrt{\dfrac{m_0m_1}{1+\tau^2}}\\ \boldsymbol{\omega}_0=2\delta\tau\sqrt{\dfrac{m_0m_1}{1+\tau^2}}\boldsymbol{l}\ ,\ \tau=\tan(\dfrac{\|\boldsymbol{x}_1-\boldsymbol{x}_0\|_2}{2\delta})\end{cases}$$

Peng, Zhou, ... Nature Methods 2026 Peng, …, Zhou. ICLR 2026 Peng*, **Chen***, Zhou. NeurIPS 2026

Peng, Zhou, ... Nature Methods 2026  Peng, …, Zhou. ICLR 2026  Peng*, **Chen***, Zhou. NeurIPS 2026

$$\mathrm{WFR}_\delta^2(\mu_0,\mu_1) = \inf_{\rho,g,\boldsymbol{u}} \int_0^1 \int_{\mathcal{X}} \frac{1}{2}(\|\boldsymbol{u}(\boldsymbol{x},t)\|_2^2 + \delta^2\|g(\boldsymbol{x},t)\|_2^2)\rho_t(\boldsymbol{x})\mathrm{d}\boldsymbol{x}\mathrm{d}t$$

$$\text{s.t.} \qquad \partial_t\rho + \nabla_{\boldsymbol{x}}\cdot(\rho\boldsymbol{u}) = \rho g,\ \rho_0 = \mu_0,\ \rho_1 = \mu_1,$$




## equivalent static Kantorovich

$$\mathrm{WFR}_\delta^2(\mu_0,\mu_1) = \inf_{(\gamma_0,\gamma_1)\in(\mathcal{M}_+(\mathcal{X}^2))^2} \int_{\mathcal{X}^2} \mathrm{WFR\text{-}DD}_\delta^2(\gamma_0(\boldsymbol{x},\boldsymbol{y})\delta_{\boldsymbol{x}},\gamma_1(\boldsymbol{x},\boldsymbol{y})\delta_{\boldsymbol{y}})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y},$$

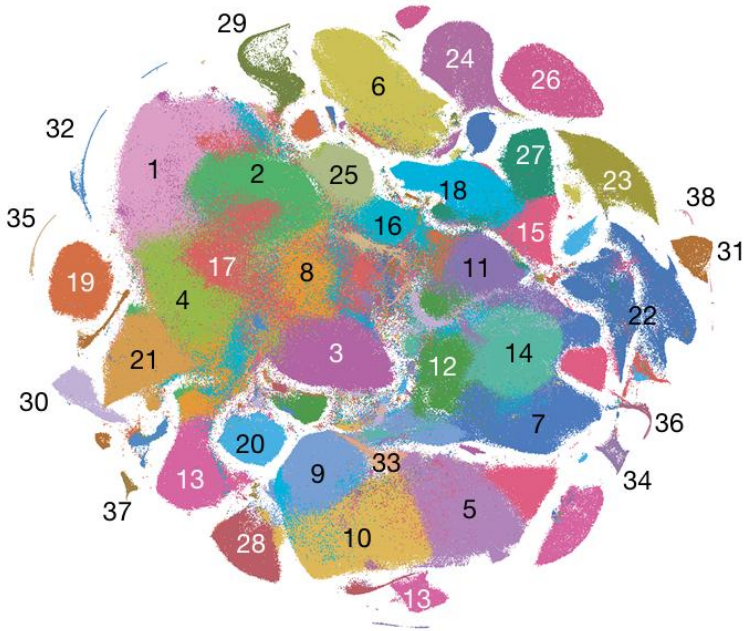
$$\Gamma(\mu_0,\mu_1) \overset{\text{def.}}{=} \left\{(\gamma_0,\gamma_1)\in\left(\mathcal{M}_+(\mathcal{X}^2)\right)^2 : \int_{\mathcal{X}}\gamma_0(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{y} = \mu_0(\boldsymbol{x}), \int_{\mathcal{X}}\gamma_1(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{x} = \mu_1(\boldsymbol{y})\right\}.$$

## optimal entropy-transport (OET)

$$\mathrm{WFR}_\delta^2(\mu_0,\mu_1) = 2\delta^2 \inf_{\gamma\in\mathcal{M}_+(\mathcal{X}^2)}\{\int_{\mathcal{X}^2} -2\ln\overline{\cos}(\frac{\|\boldsymbol{x}-\boldsymbol{y}\|_2}{2\delta})\gamma(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y}$$
$$+\mathrm{KL}(\int_{\mathcal{X}}\gamma(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{y}\|\mu_0(\boldsymbol{x})) + \mathrm{KL}(\int_{\mathcal{X}}\gamma(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{x}\|\mu_1(\boldsymbol{y}))\}$$

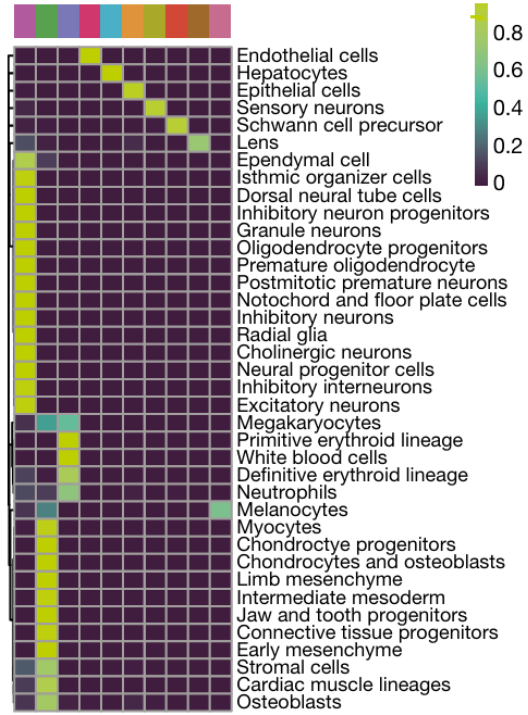
$$\gamma_0(\boldsymbol{x},\boldsymbol{y}) = \frac{\gamma(\boldsymbol{x},\boldsymbol{y})}{\int_{\mathcal{X}}\gamma(\boldsymbol{x},\boldsymbol{z})\mathrm{d}\boldsymbol{z}}\mu_0(\boldsymbol{x})$$
$$\gamma_1(\boldsymbol{x},\boldsymbol{y}) = \frac{\gamma(\boldsymbol{x},\boldsymbol{y})}{\int_{\mathcal{X}}\gamma(\boldsymbol{z},\boldsymbol{y})\mathrm{d}\boldsymbol{z}}\mu_1(\boldsymbol{y})$$

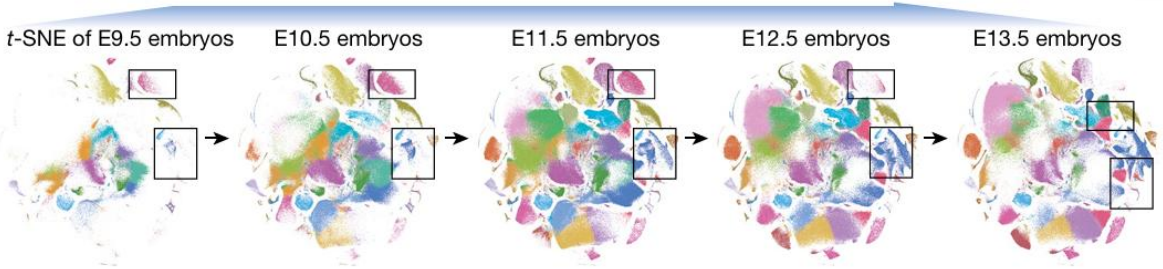
# Toward atlas-scale single-cell datasets



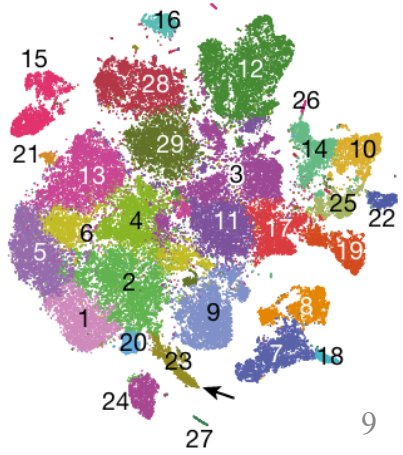

MOCA dataset annotation:

- Cell number: 1.3 million cells
- Major cell type: 38
- Sub cell type: 655



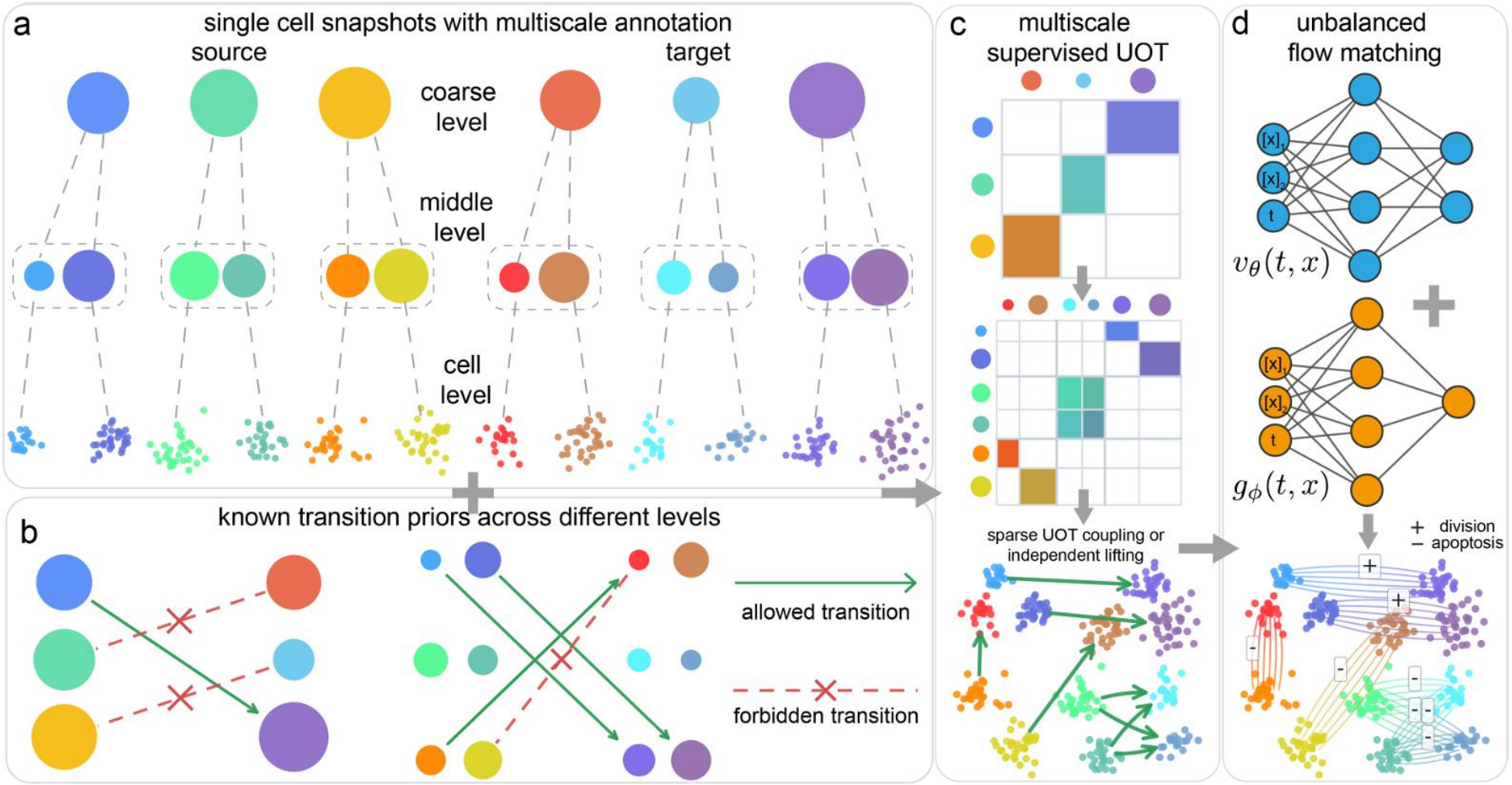




1–Keratinocyte (*Col19a1*$^+$)
2–Epidermal stem cells (*Meis1*$^+$)
3–Branchial arch ectodermal cells
4–Epidermal progenitors (*Igfbp2*$^+$)
5–Keratinocyte (*Brinp1*$^+$)
6–Keratinocyte (*Krt1*$^+$)
7–Otic vesicle epithelium
8–Otic sensory epithelium
9–Pericardium
10–Intestinal epithelium
11–Second branchial arch epithelium
12–Olfactory epithelium
13–Hair follicle stem cell
14–Intestinal stem cells
15–Renal epithelium
16–Retina epithelium
17–First branchial arch epithelium
18–Utricle and saccule epithelium
19–Lung epithelium
20–Suface ectoderm
21–Epidermal stem cells (*Lgr6*$^+$)
22–Endocrine cells
23–Apical ectodermal ridge (AER)
24–Urothelium
25–Stomach epithelium
26–Intestine epithelium (*Car3*$^+$)
27–Primordial germ cells
28–Doublets
29–Doublets




Cao et al. Nature 2019

# Multiscale Supervised Unbalanced Optimal Transport Flow Matching

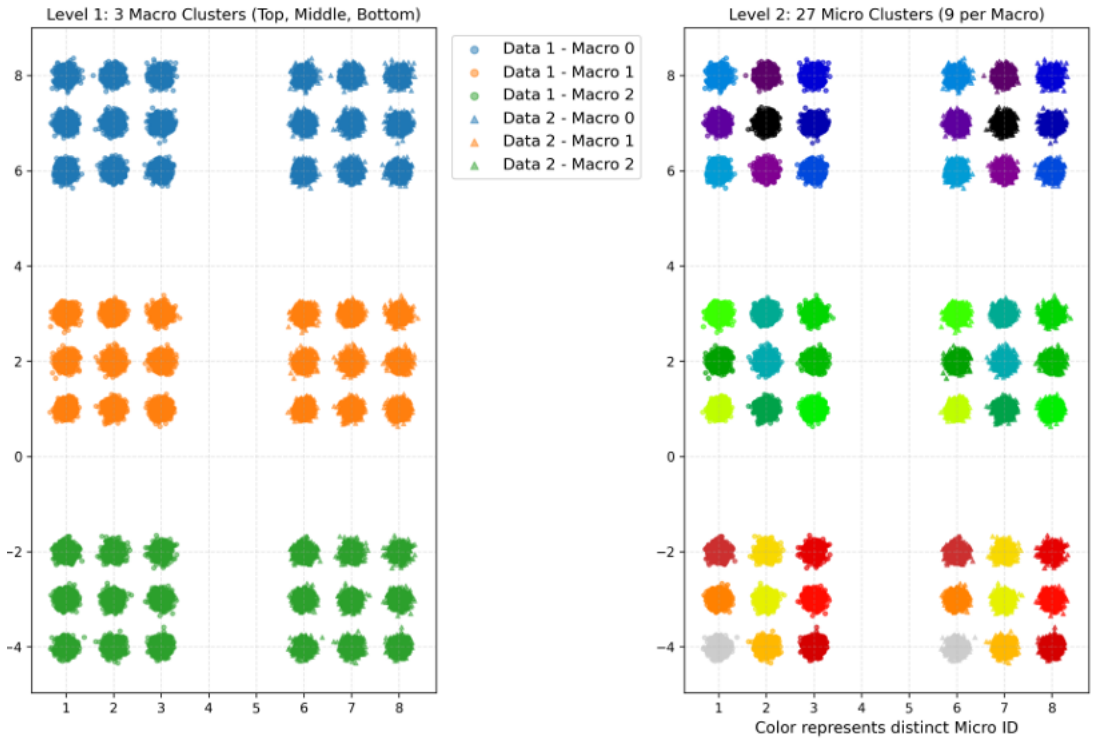



Peng*, **Chen***, Zhou. NeurIPS 2026

Peng*, **Chen***, Zhou. NeurIPS 2026

Table 1: **Coupling accuracy on the synthetic multiscale dataset (Q1).**

| Method | Obj. Gap (%) ↓ | Point Acc. ↑ | Micro Acc. ↑ | Macro Acc. ↑ | Time (s) ↓ |
|---|---|---|---|---|---|
| Ground Truth | 0.00 | 1.000 | 1.000 | 1.000 | – |
| Exact OT | 2.35e-5 | 0.268 | 0.510 | **1.000** | 89.91 |
| Mini-batch OT [33] | 1.50e-1 | 0.002 | 0.328 | 0.988 | 8.86 |
| Low-rank OT [31] | 1.35e+1 | 0.000 | 0.076 | 0.695 | 151.92 |
| MUST-FM (Sparse) | **1.78e-14** | **0.999** | **1.000** | **1.000** | 10.50 |
| MUST-FM (Independent) | 3.99e-2 | 0.001 | **1.000** | **1.000** | **0.025** |



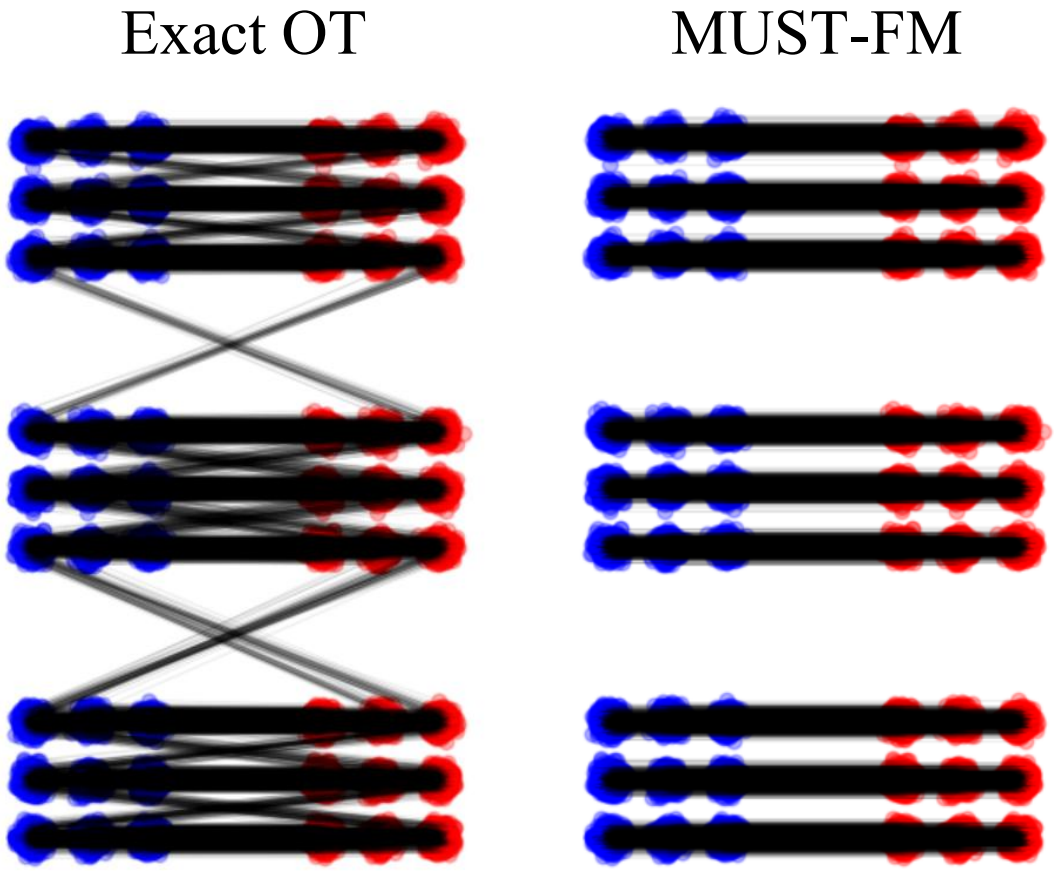



Table 3: **Leave-one-out interpolation on MOCA dataset [34] (Q3).** We report $W_1$ (↓) at each held-out time point.

| Method | E10.5 | E11.5 | E12.5 | Average |
|---|---|---|---|---|
| MUST-FM w/o Priors | 1.370 | 1.429 | **1.252** | 1.350 |
| MUST-FM w/ Priors | **1.326** | **1.364** | 1.318 | **1.336** |

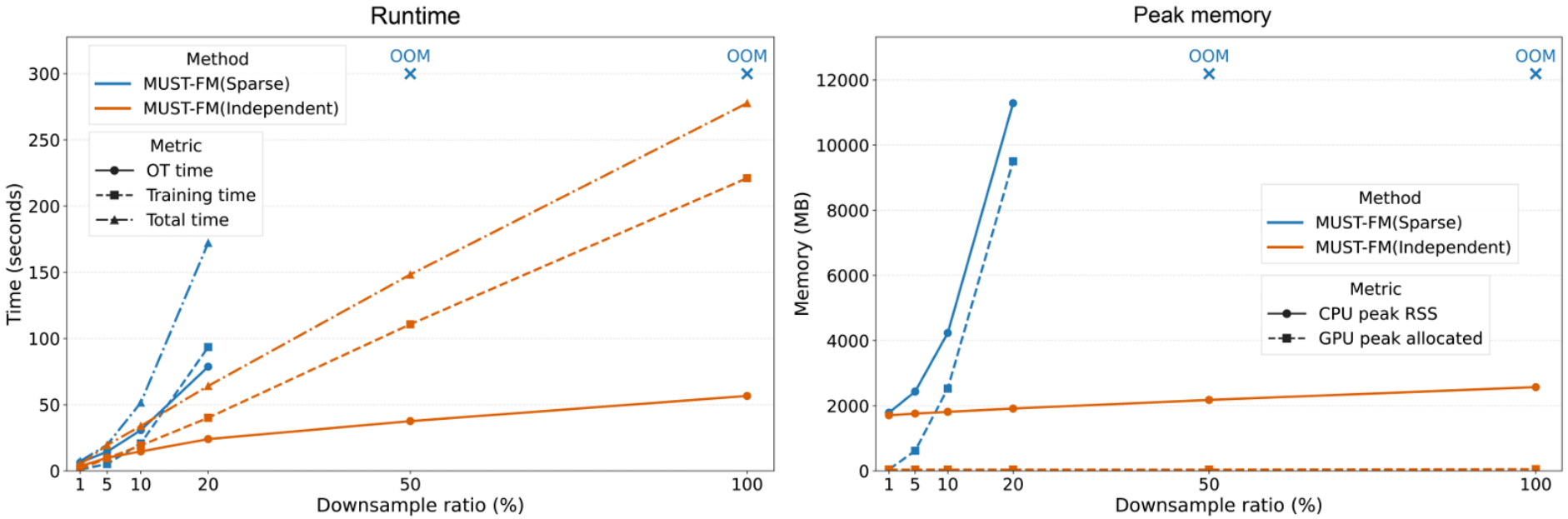
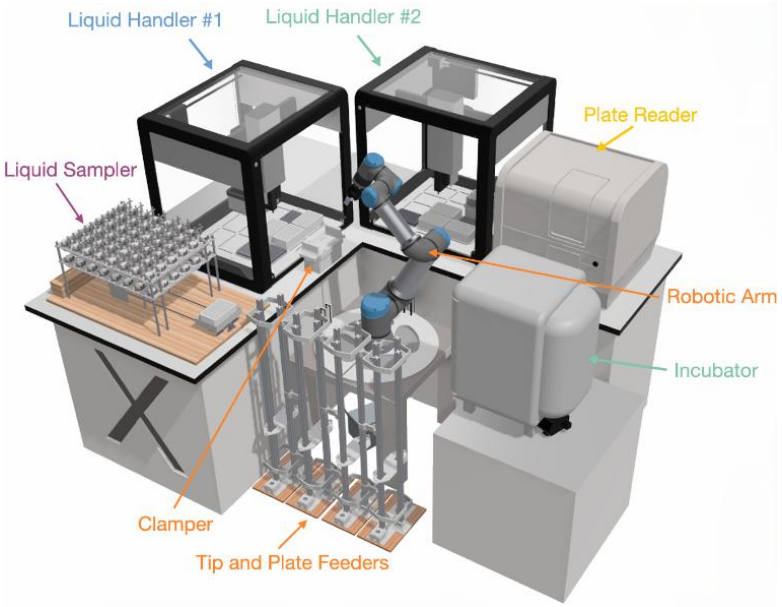



Figure 2: **Scalability on MOCA [34] dataset.** Runtime and memory usage of MUST-FM under different subsampling ratios.

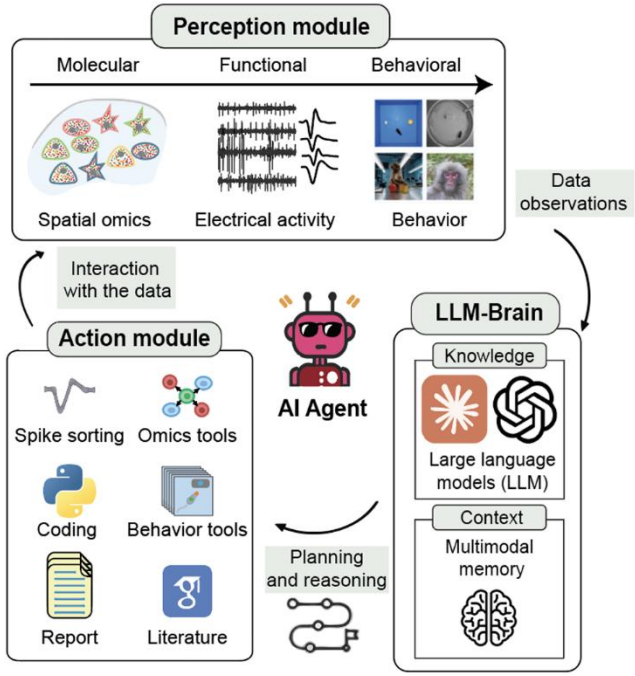
# Future plan




Theoretical modeling

High-throughput data

Computational methods




Closed-loop validation